

Aufgabensammlung Modul 114

Gedruckt am : 19.06.2026

Aufgabe 1: Encoding mit Spass

Note that unlike other Markdown renderers (notably Jupyter and GitHub), lists in Quarto require an entire blank line above the list. Otherwise the list will not be rendered in list form, rather it will all appear as normal text along a single line.

Titel 2

Right	Left	Default	Center
12	12	12	12
123	123	123	123
1	1	1	1

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sagittis posuere ligula sit amet lacinia. Duis dignissim pellentesque magna, rhoncus congue sapien finibus mollis. Ut eu sem laoreet, vehicula ipsum in, convallis erat. Vestibulum magna sem, blandit pulvinar augue sit amet, auctor malesuada sapien. Nullam faucibus leo eget eros hendrerit, non laoreet ipsum lacinia. Curabitur cursus diam elit, non tempus ante volutpat a. Quisque hendrerit blandit purus non fringilla. Integer sit amet elit viverra ante dapibus semper. Vestibulum viverra rutrum enim, at luctus enim posuere eu. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus.

code

Bilder



Abbildung 1: Elephant

page 2

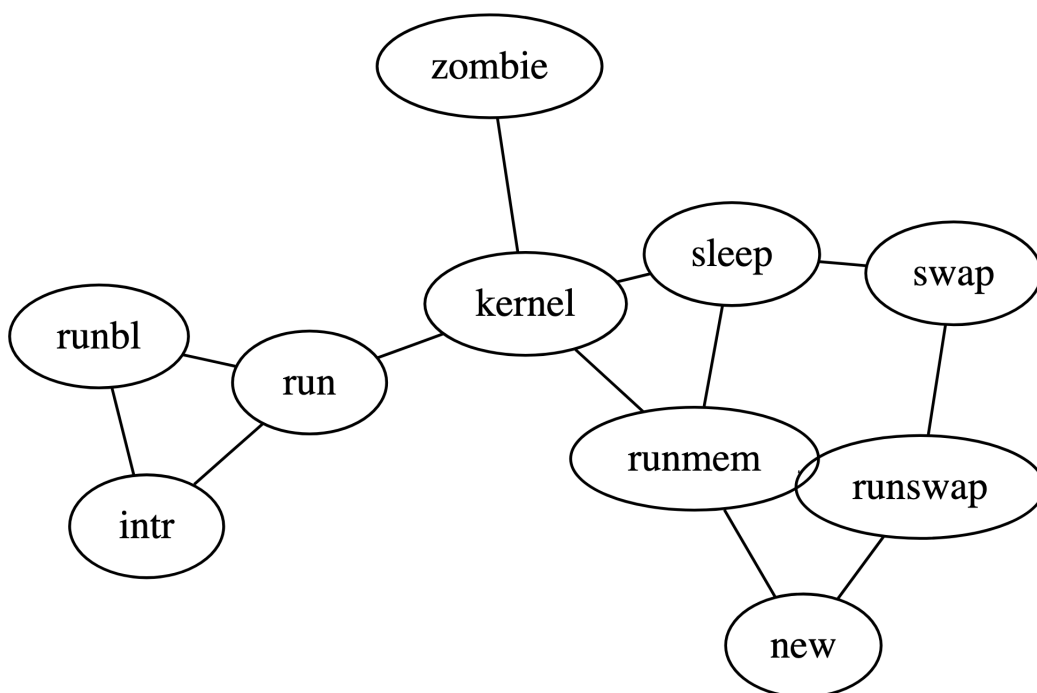
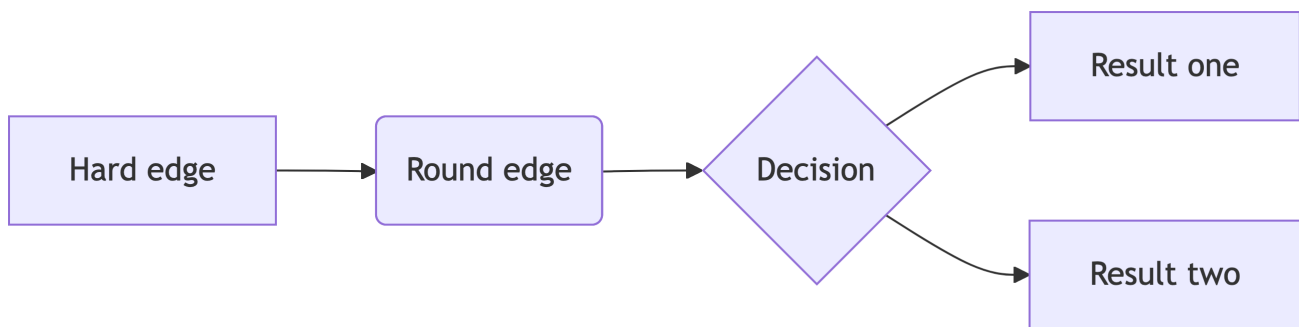
Fussnoten

Here is a footnote reference,¹ and another.²

Spezialzeichen

→ Pfeil

Diagramme



Titel 3

! Important

Note that there are five types of callouts, including: `note`, `warning`, `important`, `tip`, and `caution`.

¹Here is the footnote.

²Here's one with multiple blocks.

⚠ Warning

Note that there are five types of callouts, including: `note`, `warning`, `important`, `tip`, and `caution`.

💡 Tip

Note that there are five types of callouts, including: `note`, `warning`, `important`, `tip`, and `caution`.

💡 Tip with Title

This is a callout with a title.

Titel 2**i Aufgabe 1: Das ist ein Text**

Note that there are five types of callouts, including: `note`, `warning`, `important`, `tip`, and `caution`.

💡 Tip with Title

This is an example of a callout with a title.

🔥 Expand To Learn About Collapse

This is an example of a ‘folded’ caution callout that can be expanded by the user. You can use `collapse="true"` to collapse it by default or `collapse="false"` to make a collapsible callout that is expanded by default.

Aufgabe 2: Das sind andere Übungen

Note that unlike other Markdown renderers (notably Jupyter and GitHub), lists in Quarto require an entire blank line above the list. Otherwise the list will not be rendered in list form, rather it will all appear as normal text along a single line.

Titel 2

Right	Left	Default	Center
12	12	12	12
123	123	123	123
1	1	1	1

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sagittis posuere ligula sit amet lacinia. Duis dignissim pellentesque magna, rhoncus congue sapien finibus mollis. Ut eu sem laoreet, vehicula ipsum in, convallis erat. Vestibulum magna sem, blandit pulvinar augue sit amet, auctor malesuada sapien. Nullam faucibus leo eget eros hendrerit, non laoreet ipsum lacinia. Curabitur cursus diam elit, non tempus ante volutpat a. Quisque hendrerit blandit purus non fringilla. Integer sit amet elit viverra ante dapibus semper. Vestibulum viverra rutrum enim, at luctus enim posuere eu. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus.

code

Bilder



Abbildung 2: Elephant

page 1

Aufgabe 3: Und noch andere Sachen

Probability Basics

The probability of an event A is denoted as $P(A)$ and satisfies:

$$0 \leq P(A) \leq 1$$

[1]

For two events A and B :

The **median** is the middle value when data is sorted.

Mean (Average)

Definition 3.1

The **mean** of a dataset is the sum of all values divided by the number of values. It represents the central tendency of the data and is sensitive to outliers.

For a sample: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

Variance and Standard Deviation

The sample variance is:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad [2]$$

And the standard deviation is simply $s = \sqrt{s^2}$.

Aufgabe 4: Hypothesis Testing

The Framework

1. State null hypothesis H_0 and alternative H_A
2. Choose significance level α (typically 0.05)
3. Calculate test statistic
4. Make decision based on p-value

t-Test Example

For a one-sample t-test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad [3]$$

where μ_0 is the hypothesized population mean.

t-Distribution

Theorie 4.1

Under the null hypothesis, if the population is normally distributed, the test statistic follows a t-distribution with $n - 1$ degrees of freedom.

Aufgabe 5: Conclusion

This covers the basics of statistical analysis. Next lecture will focus on regression analysis and correlation.

Using Colored Boxes

You can use the colored theorem boxes by inserting raw Typst code:

Central Limit Theorem

Definition 5.1

The distribution of sample means approximates a normal distribution as the sample size becomes larger, regardless of the population's distribution shape.

Law of Large Numbers

Theorie 5.1

As the number of trials increases, the sample mean converges to the expected value.

Sample Mean Calculation

Beispiel 5.1

For the dataset {2, 4, 6, 8, 10}, the mean is:

$$\bar{x} = \frac{2 + 4 + 6 + 8 + 10}{5} = 6 \quad [4]$$

Important Note

Notiz 5.1

Always check your assumptions before applying statistical tests.

Warning

Wichtig 5.1

Sample size must be at least 30 for the Central Limit Theorem to apply reliably.

Available box types: - `#definition[]` - Olive colored (for definitions) - `#theorem[]` - Navy colored (for theorems) - `#proposition[]` - Maroon colored (for propositions) - `#example[]` - Purple colored (for examples) - `#note[]` - Blue colored (for notes) - `#attention[]` - Red colored (for warnings/attention) - `#quote[]` - Black colored (for quotes) - `#hypothesis[]` - Orange colored (for hypotheses)

End of lecture notes